PARISHRAM 2.0

CLASS 12TH 2026

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Electric Charges and Fields

PHYSICS **LECTURE-4**

BY-DEV AGNANI SIR



TOPICS TO BE COVERED

1. Electric field
2. Electric field due to a point charge
3. Electric field due to a group of charges → इलवा est.

Recap

1. Vector form of coulomb's law
2. Principle of superposition
3. Forces between multiple charges
4. Electric field



$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

$$\vec{F}_{21} = \frac{kQ_1Q_2}{|\vec{r}_{21}|^3}\vec{r}_{21}$$

$$= \frac{kQ_1Q_2}{|\vec{r}_{21}|^2} \times \frac{\vec{r}_{21}}{|\vec{r}_{21}|}$$

$$\Rightarrow \boxed{\vec{F}_{21} = \frac{kQ_1Q_2}{|\vec{r}_{21}|^2}\hat{r}_{21}}$$

Parishram 2.0 batch students to Dev sir when he kept on solving that equilateral triangle question:

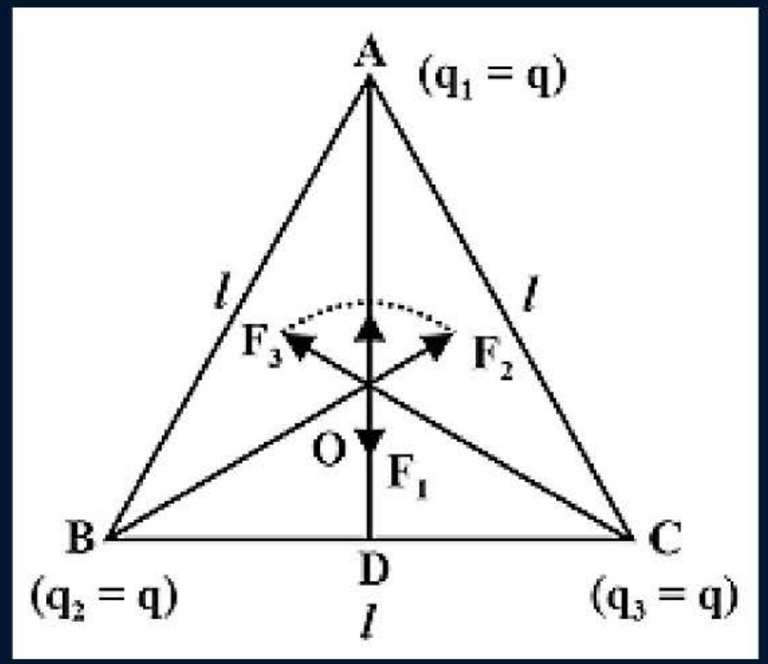


QUESTION

Consider three charges q_1, q_2, q_3 each equal to q at the vertices of an equilateral triangle of side l. What is the force on a charge Q (with the same sign as q) placed at the centroid of the triangle, as shown in Figure.

Centroid of a Δ :-

Centre of a Δ where its centre of mass lies.

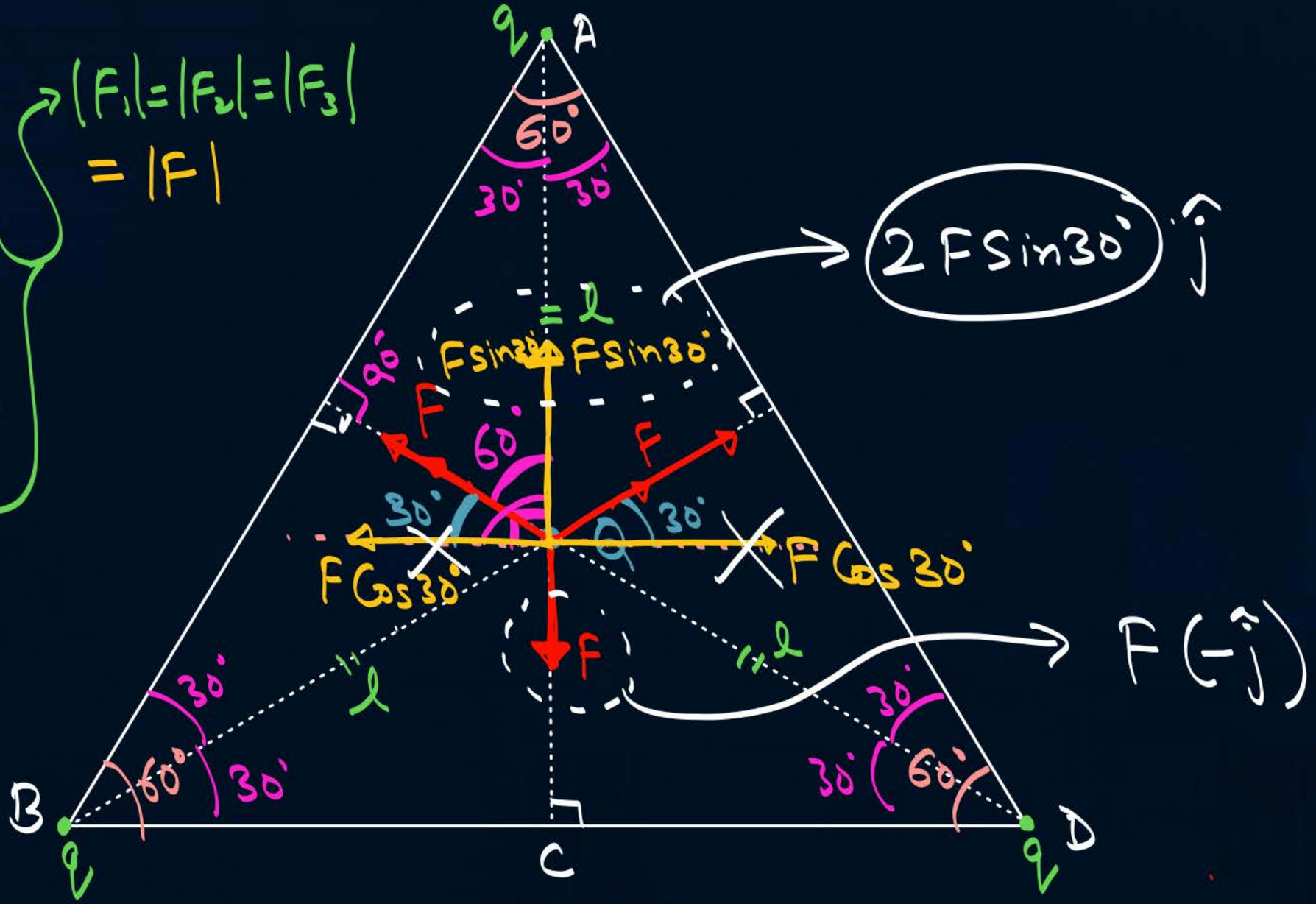


$$|F_1| = \frac{kq^2}{l^2}$$

$$|F_2| = \frac{kq^2}{l^2}$$

$$|F_3| = \frac{kq^2}{l^2}$$

$$\Rightarrow |F_1| = |F_2| = |F_3| = |F|$$



$$\rightarrow 2F\sin 30^\circ\,\hat{j}$$

$$\rightarrow F(-\hat{j})$$

Net Force on charge Q placed at the centroid :-

$$(\vec{F}_{net})_Q = (2F\sin 30^\circ)\hat{j} + (F)(-\hat{j})$$

$$= \left\{\left(2F \times \frac{1}{2}\right) - (F)\right\}\hat{j}$$

$$\boxed{(\vec{F}_{net})_Q = 0}$$

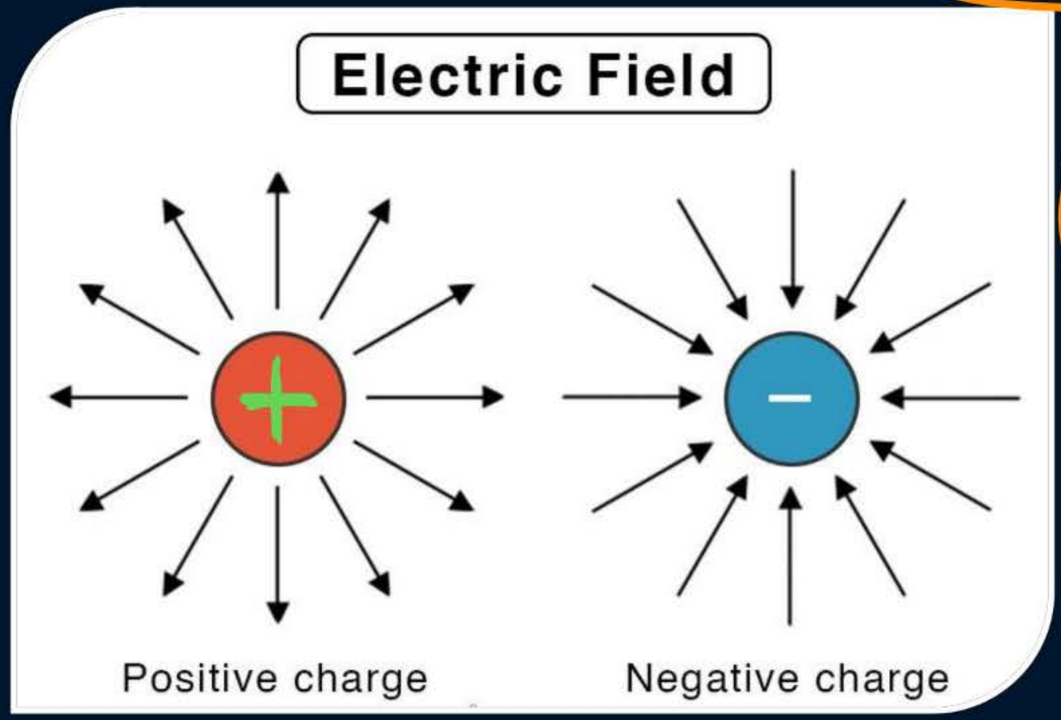
Electric Field

OR Electric field intensity OR Intensity

- An electric field is the region around a charged object in which another charge experiences a force.

$$E = \frac{F}{q}$$

Scalar form of Electric field.



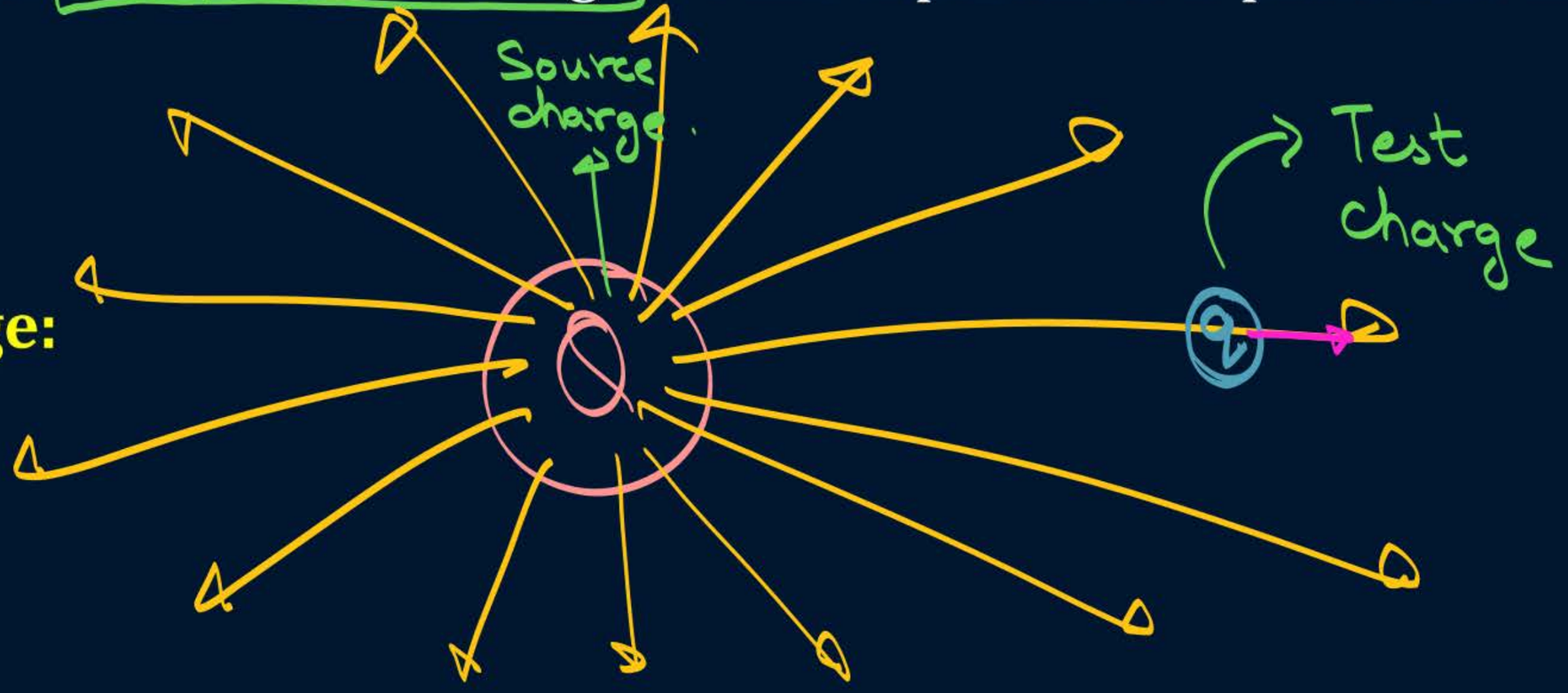
+ve charge → Towards outside

-ve charge → towards inside.

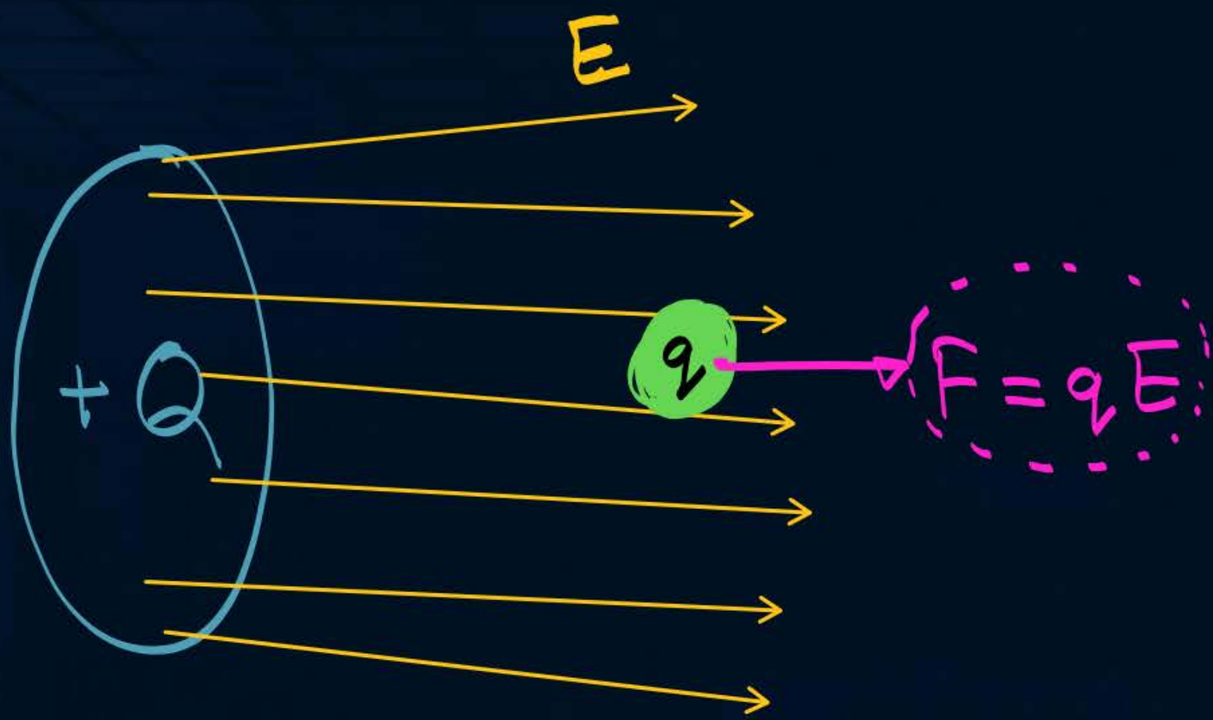
Electric Field

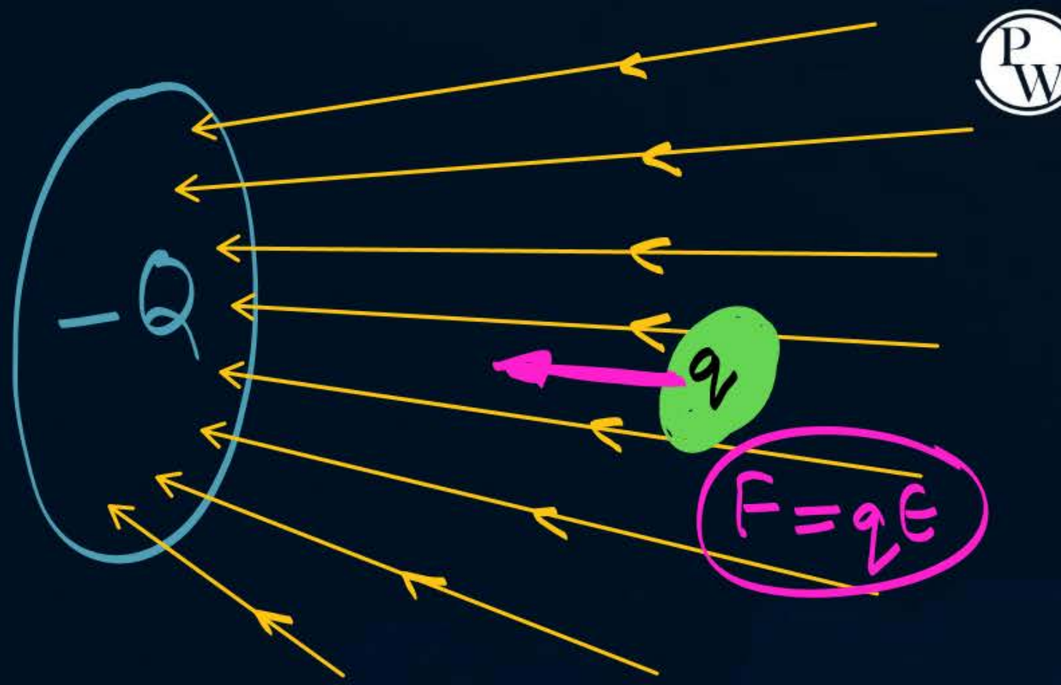
Unit → 1

- The electric field due to a charge Q at a point in space may be defined as the force that a unit positive charge would experience if placed at that point.



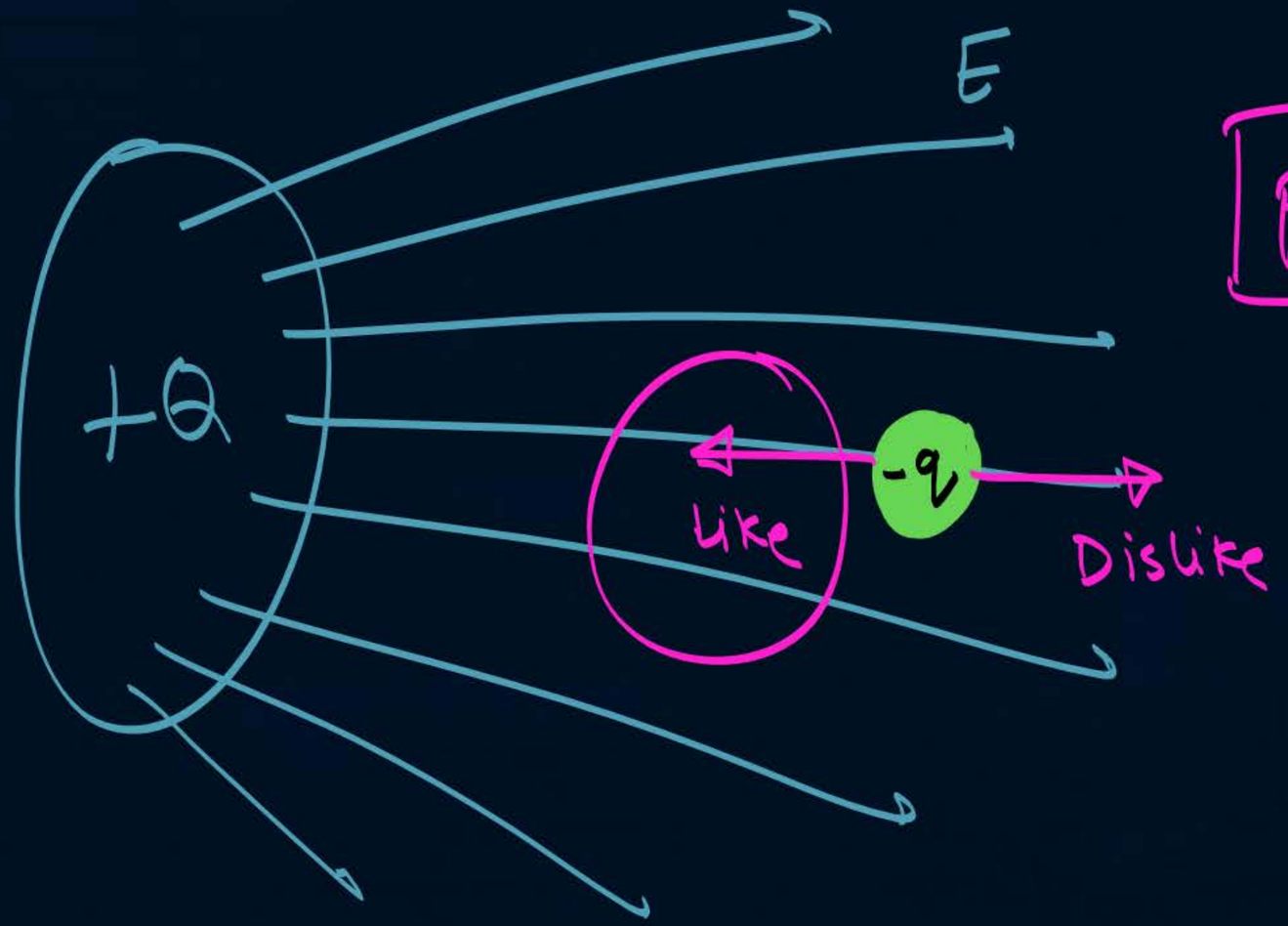
- **Source Charge:**
- **Test Charge:**





$$E = \frac{F}{q}$$

$$\Rightarrow F = Eq$$



$$F = -qE$$

Unit of Electric Field

→ NC^{-1}
→ Vm^{-1}

जब तक Formula नहीं पता, तब तक unit की बात नहीं कर सकते।

$E = \frac{f}{q}$

→ unit : Newton (N)

→ unit : Coulomb (C)

Newton/Coulomb

$= NC^{-1}$

Electric Potential

$V = Ed$

→ Volt (V)

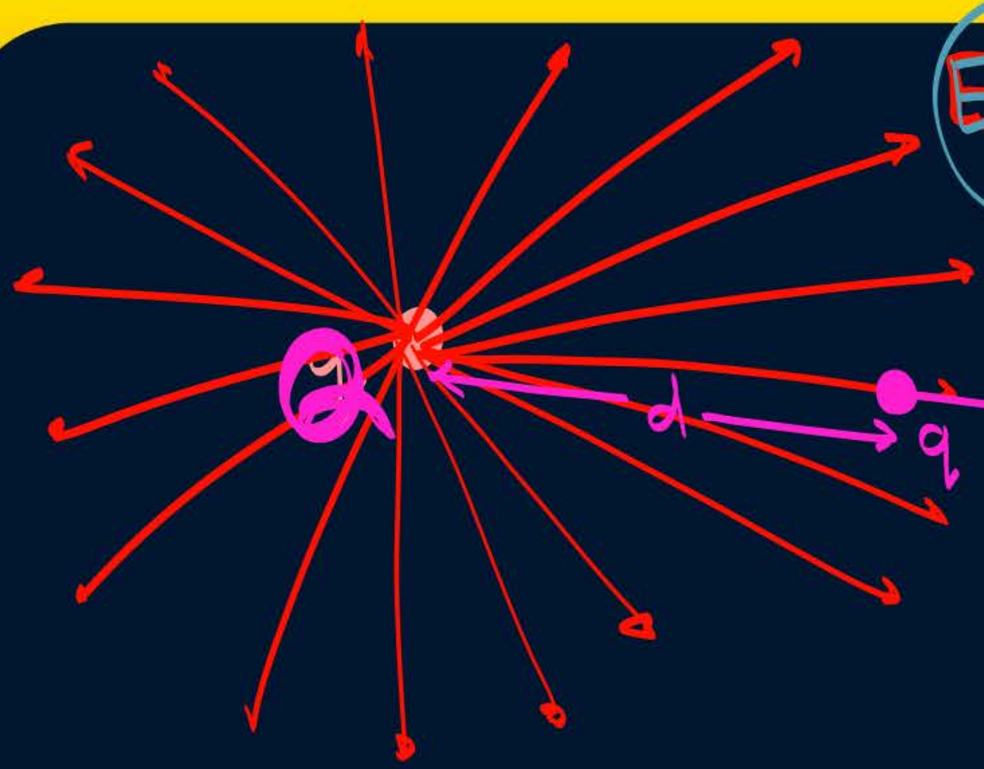
→ m

$\Rightarrow E = \frac{V}{d}$

Volt/meter

$\Rightarrow Vm^{-1}$

Electric Field due to a point charge



$$E = \frac{F}{q} \Rightarrow E = \frac{\frac{kQq}{d^2}}{q}$$

$$F = \frac{kQq}{d^2}$$

$$\Rightarrow \boxed{E = \frac{kQ}{d^2}}$$

Vector form of Electric Field :-

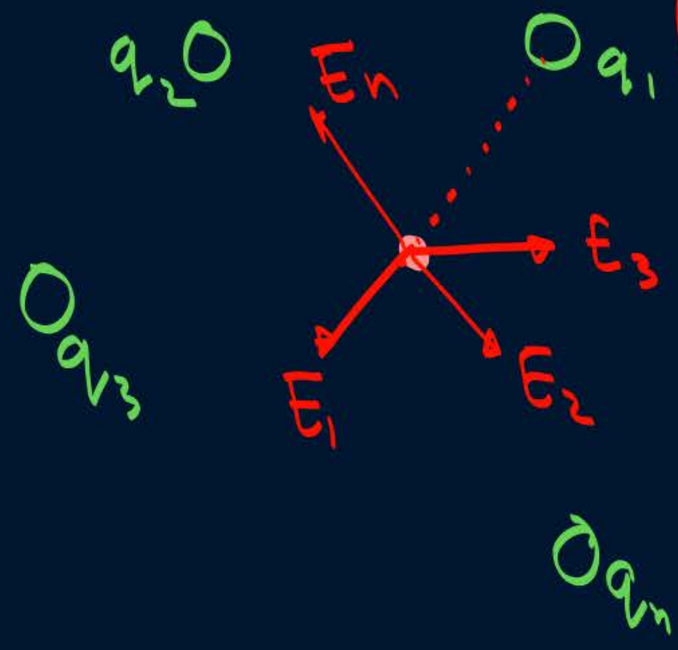
$$\vec{E} = \frac{\vec{F}}{q} \Rightarrow \vec{E} = \frac{\frac{kQq}{|\vec{d}|^3}\vec{d}}{q} \Rightarrow \boxed{\vec{E} = \frac{kQ}{|\vec{d}|^3}\vec{d}}$$

$$\Leftarrow \vec{E} = \frac{kQ}{|\vec{d}|^2}\frac{\vec{d}}{|\vec{d}|} \Leftarrow \boxed{\vec{E} = \frac{kQ}{|\vec{d}|^2}\hat{d}}$$

Electric Field due to a group of charges

- If instead of a single charge, let the electric field is produced by n number of charges, using the principle of superposition resultant electric field intensity

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \cdots$$



→ Net Electric field on a charge is the vector sum of all the Electric fields under whose influence the charge is present.

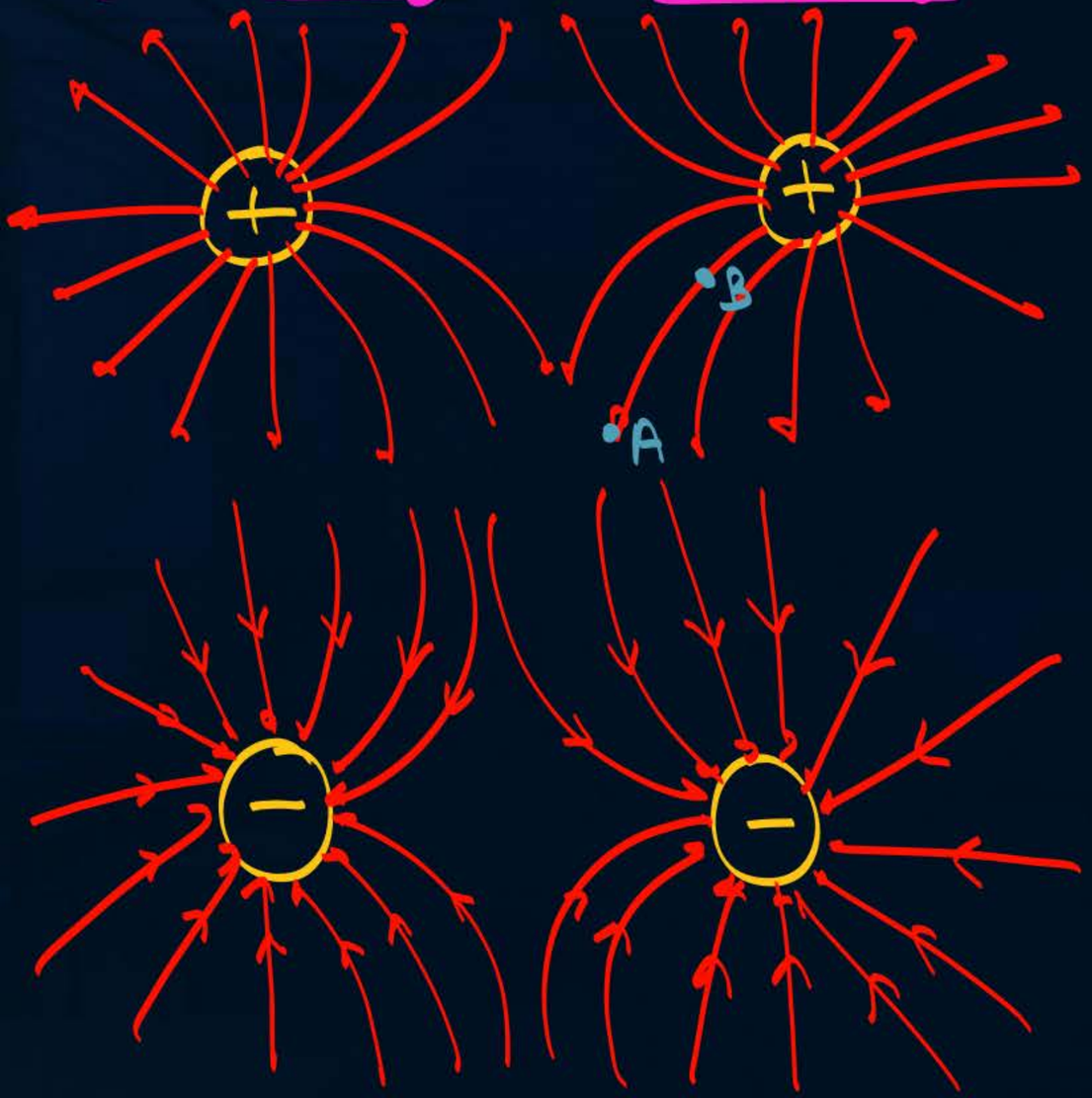
Null Points

Knowledge.

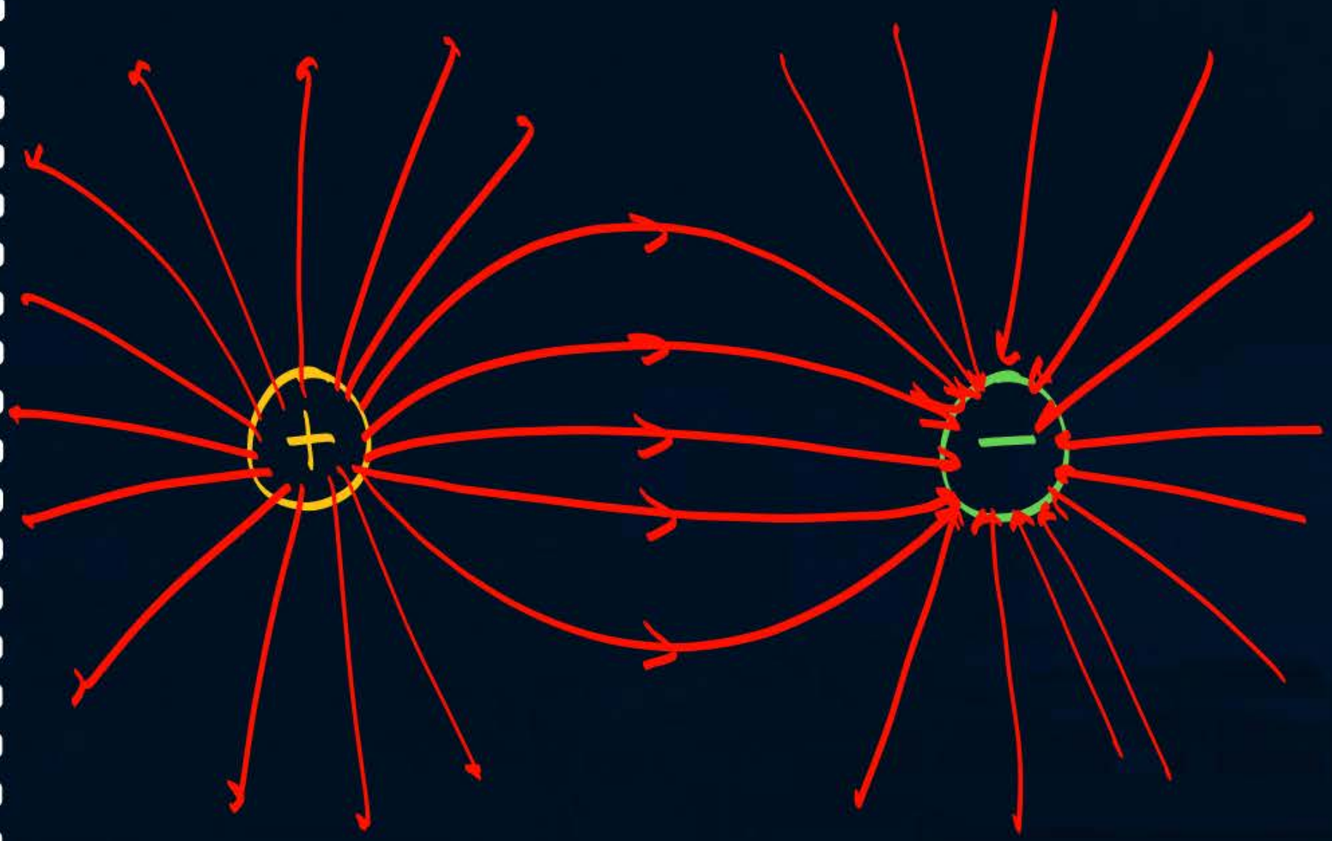
- Two charges q_1 and q_2 are separated by a distance 'd'. Then the point of zero intensity (null point) lies at a distance of $x = \frac{d}{\sqrt{\frac{q_2}{q_1}} \pm 1}$ from q_1

  +ve sign for like charges -ve sign for unlike charges.

Both charges are like charges:



Both charges are unlike charges:



QUESTION

A charge of 2×10^{-6} C is placed in an electric field of strength 5×10^{4} N/C. What is the force acting on the charge?

A. 0.1 N

B. 0.5 N

C. 1.0 N

D. 2.0 N

$E = \frac{F}{q} \Rightarrow F = Eq$

$\Rightarrow F = \left(5 \times 10^{4} \frac{N}{C}\right) \times (2 \times 10^{-6} C)$

$\Rightarrow F = 10 \times 10^{-2}$ N

$\Rightarrow F = 10^{-1}$ N

$\Rightarrow \boxed{F = 0.1 N}$.



QUESTION

What is the electric field at a point 0.2 m away from a point charge of 3×10^{-6} C ? ($k = 9 \times 10^9$ Nm^2/C^2)

A 6.75×10^5 N/C

B 3.75×10^5 N/C

C 2.25×10^6 N/C

D 1.5×10^6 N/C

$$E = \frac{kQ}{d^2} = \frac{(9 \times 10^9)(3 \times 10^{-6})}{(0.04)}$$

$$= \frac{27}{0.04} \times 10^3 \times 10^2 = \frac{27}{4} \times 10^5$$

$$= 6.75 \times 10^5 \text{ N/C}$$



QUESTION

What is the direction of the electric field due to a positive point charge?

A Inward, toward the charge

B Circular around the charge

C Outward, away from the charge

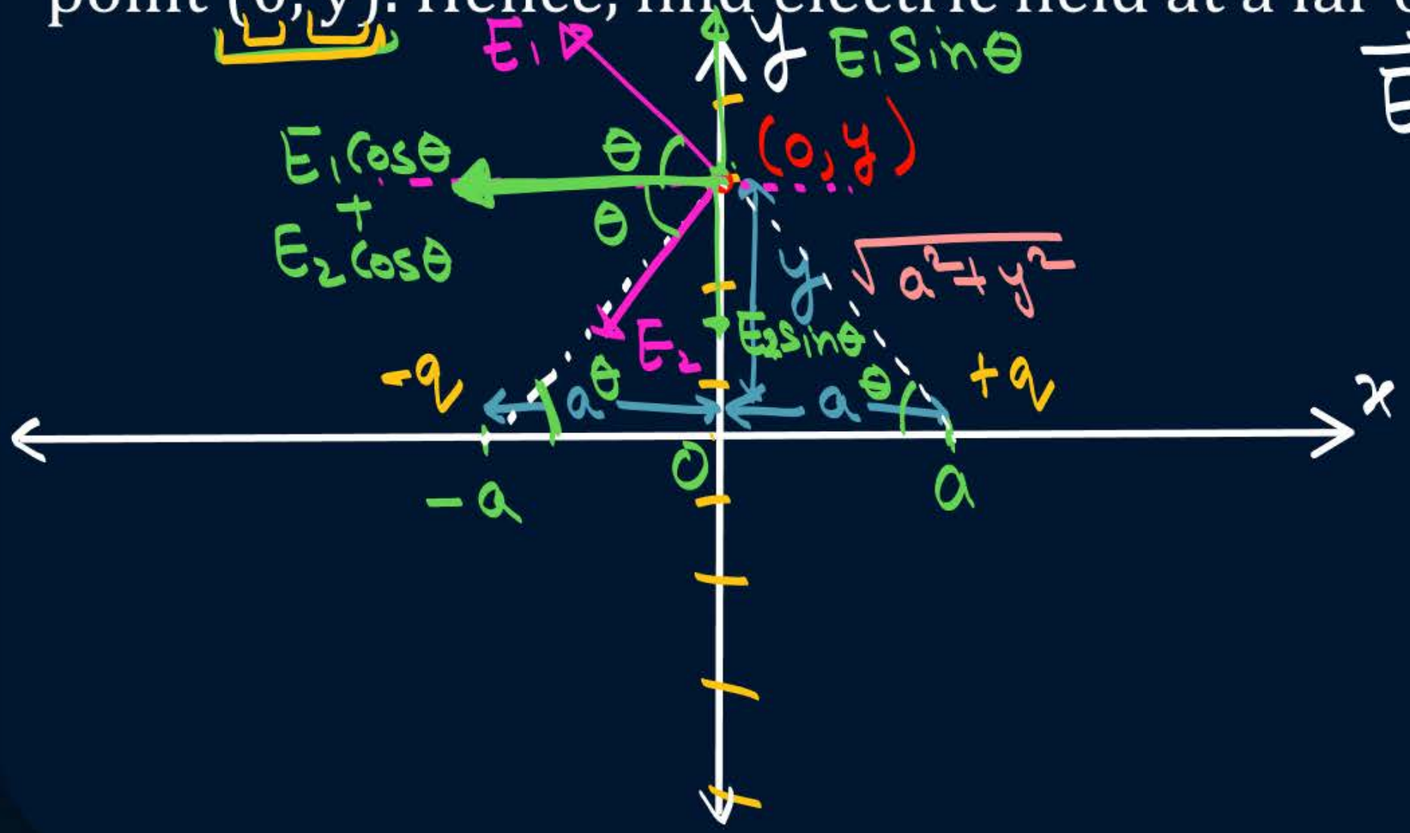
D None of these



QUESTION

CBSE 2025 | 55/4/3 Set – 3

Two point charges +q and -q are held at (a, 0) and (-a, 0) in x-y plane. Obtain an expression for the net electric field due to the charges at a point (0, y). Hence, find electric field at a far off point (y ≫ a).



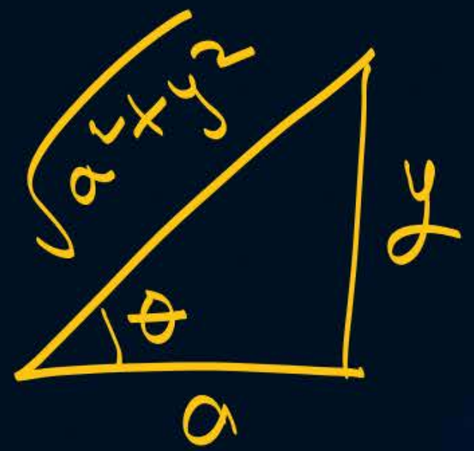
$$\vec{E}_{net} = \vec{E}_1 + \vec{E}_2$$

$$E_1 = \frac{kq}{(\sqrt{a^2+y^2})^2} = \frac{kq}{a^2+y^2} = E$$

$$E_2 = \frac{k(-q)}{(\sqrt{a^2+y^2})^2} = \frac{-kq}{a^2+y^2} = -E$$

$$\vec{E}_{net} = (2E\cos\theta)(-\hat{i})$$

$$= 2\frac{kq}{(a^2+y^2)} \times \frac{a}{\sqrt{a^2+y^2}}(-\hat{i})$$



$$\cos\theta = \frac{B}{H}$$

$$\Rightarrow \cos\theta = \frac{a}{\sqrt{a^2+y^2}}$$

$$\vec{E}_{net} = -\frac{2kqa}{(a^2+y^2)^{3/2}}\hat{i}$$

Given that $y >> a$

$\Rightarrow y^2 >> >>> a^2$

$\Rightarrow$ $$\boxed{\vec{E}_{net} = \frac{-2Kqa}{y^3}\hat{i}}$$

QUESTION

NCERT

HOMEWORK

What is the force between two small charged spheres having charges of 2 $\times 10^{-7}$ C and 3 $\times$ 10-7 C placed 30 cm apart in air?

QUESTION

NCERT

HOMEWORK

Two point charges $q_A = 3\mu C$ and $q_B = -3\mu C$ are located 20 cm apart in vacuum.
(a) What is the electric field at the midpoint O of the line AB joining the two charges?
(b) If a negative test charge of magnitude $1.5 \times 10^{-9} C$ is placed at this point, what is the force experienced by the test charge?

QUESTION

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HOMEWORK.

The magnitude of electric force F exerted by a charged particle on another charged particle is 2.4×10^{-8} N. If the distance between them becomes half of its previous value, then the magnitude of the exerted force will be

A 1.2×10^{-8} N

B 4.8×10^{-8} N

C 9.6×10^{-8} N

D 19.2×10^{-8} N



THANK YOU

